

DATA266 Lab 1

LLM pretraining, sentiment classification, and CycleGAN style transfer

Team	18
Members	Aswin John and Chaitanya Uppalapati
Repository	github.com/ChaitanyaUppalapati/Lab_Pair_18
Report date	October 6, 2026

Team-selected results

Task 1: Chaitanya's 10.8M-parameter GPT (validation CE 0.5097). Task 2: Aswin's TextCNN (92.52% accuracy, 2.59M parameters), selected as the lead Task 2 model. Task 3: Chaitanya's final CycleGAN submission (FID 96.0061, MiFID 0.3998, public score -48.2029, team rank 15th).

Purpose

This joint report compares every independently trained member model, selects the strongest team result for each task, and traces all claims to repository evidence. It combines quantitative evaluation with loss curves, qualitative outputs, failure cases, robustness analysis, hardware disclosure, and reproducibility notes.

Integrity note

No metric, human rating, private leaderboard score, or rank is inferred when supporting evidence is absent. The private leaderboard fields are therefore left open until the competition closes.

Team ownership statement

Aswin and Chaitanya each maintained named folders and independently ran all three tasks. Aswin trained the compact character GPT on an A100, the three Yelp classifiers on Apple M4/MPS, and the V2 CycleGAN on an RTX 4090, and preserved the executed notebooks, checkpoints, predictions, metrics, and plots. Chaitanya designed and ran the larger GPT, the fastText/HAN/Transformer sentiment lineup, and the iterative CycleGAN experiments on an RTX 4090. Implementation and debugging used AI assistance: Chaitanya's code was written with an AI assistant from Chaitanya's design decisions, and Task 3 decisions from run 5 onward are marked as AI-assisted in DESIGN_LOG.md. Aswin's code was written with AI coding assistants (OpenAI Codex and Claude) from Aswin's design decisions; Aswin chose the architectures and hyperparameters, ran the training, and reviewed the outputs. This report was compiled with AI assistance from the repository evidence. Both members rated the 30-sample Task 3 human audit. Both members own the submitted artifacts and remain responsible for explaining their individual choices and results. This report was synthesized from the recorded evidence and should be reviewed by both members before submission.

Evaluation and selection policy

- Use the shared 5,000-review test split for cross-member Task 2 comparisons.
- Use the instructor's first-300-images evaluation notebook for the official Task 3 FID/MiFID comparison.
- Treat protocol differences as confounds, and do not call a difference architectural when preprocessing or data exposure also differs.
- Select by the task's primary held-out or official score; use efficiency and qualitative behavior as secondary criteria.

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1. Task 1 - Character-level GPT from scratch

Both models implement causal decoder-only Transformers without prebuilt Transformer or attention modules. The models differ substantially in capacity, context length, cleaning, and how much text is exposed per epoch, so the comparison measures complete pipelines rather than a single controlled ablation.

1.1 Architecture, data, and training

Item	Chaitanya full_run01	Aswin full_run01
Data	100K train / 10K validation stories; cleaned, exact-deduplicated; sliding windows, stride 128	100K train / 10K validation stories; one seeded crop or padded sequence per story
Vocabulary	85 characters incl. EOS/UNK; mojibake stories removed	111 entries incl. BOS/EOS/PAD/UNK; raw character coverage retained
Architecture	6 pre-LN blocks; d_model 384; 6 heads x 64; FFN 1536; learned positions	5 pre-LN blocks; d_model 240; 6 heads x 40; FFN 960; learned positions
Context / dropout	256 / 0.10	160 / 0.12
Optimizer	AdamW, lr 1e-3, wd 0.1, betas 0.9/0.99	AdamW, lr 2.5e-4, wd 0.1
Schedule	1,000-step warm-up; cosine to 1e-4	5% warm-up; cosine to 10% of peak
Batch / epochs	64 / 10	48 / 12
Hardware	RTX 4090, fp32	Google Colab Pro A100-SXM4-40GB

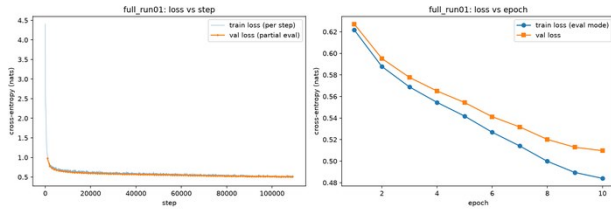
1.2 Complete comparison metrics

Metric	Chaitanya	Aswin
Training CE loss	0.4842	0.7932
Validation CE loss	0.5097	0.7582
Perplexity	1.6647	2.1345
Bits per character	0.7353	1.0939
Generalization gap	0.0255	-0.0350
Top-1 next-char accuracy	0.8358	0.7601
Distinct-1 / 2 / 3	0.3443 / 0.7785 / 0.9292	0.0167 / 0.1180 / 0.3220
Repeated 4-gram rate	0.0195	0.5115
Gradient norm max / mean	9.627 / 0.201	6.871 / 0.720
Loss spikes / NaNs	0 / 0	125 / 0
Parameters	10,811,136	3,558,960
Train tokens/s	235,385	233,568
Generation tokens/s	179.9	241.1
Peak memory	4,598 MB	1,018 MB
Training time	8,458 s	822 s

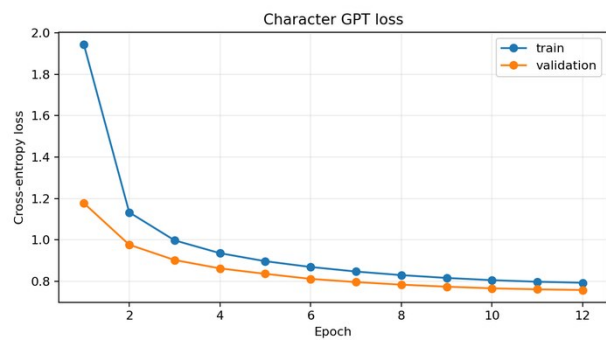
Team selection: Chaitanya full_run01

It has the lower validation loss (0.5097 vs 0.7582), lower perplexity (1.665 vs 2.134), and higher next-character accuracy (83.58% vs 76.01%). Aswin's model is the efficiency winner: it is about one third the size, trains about 10.3x faster in wall-clock time, uses about 4.5x less peak memory, and generates faster. Distinct-n values are not directly comparable because one pipeline uses word n-grams and the other character n-grams.

1.3 Training evidence



Chaitanya: step-level and epoch-level curves. Validation loss falls through epoch 10 with a small positive gap.



Aswin: loss falls through epoch 12. The lower validation curve reflects dropout being disabled during validation, not leakage.

1.4 Joint analysis

The stronger GPT benefits from roughly 3x more parameters, 60% longer context, aggressive cleaning/deduplication, and sliding-window exposure that provides roughly ten times more target characters per epoch. Its zero-spike trace and 0.0255 generalization gap show stable underfitting: validation loss was still decreasing, so additional epochs or mixed precision would likely improve it. Aswin's compact run demonstrates excellent throughput and memory efficiency, but one crop per story and a 160-character context weaken story coherence. The negative reported gap in that run mixes dropout-on training loss with dropout-off validation loss and should not be interpreted as genuine validation superiority. A controlled follow-up would standardize the split, vocabulary, metric definitions, and number of target characters seen, then ablate context length and model size separately.

1.5 Failure cases with actual generated text

Member / case	Generated excerpt	Failure and diagnosis
Chaitanya 1	"The dog and the box became best friends. They played together every day." repeats twice.	Greedy repetition loop; a frequent template becomes self-reinforcing.
Chaitanya 2	"an x-ray of te-ta-ta--cars ... clean her tea-flashers"	Temperature-1.0 character sampling creates invented words and semantic drift.
Chaitanya 3	"learnt an important lesson - that woggly river my work tasks"	A familiar moral template begins correctly but long-range syntax collapses.
Aswin 1	"The bird was so happy that he had been so much fun to see the bird" repeats to the cap.	Greedy repetition, amplified by short context and single-crop training.
Aswin 2	"a bugs ... a cloth their basket ... mojibake quotation marks"	High-temperature output exposes retained encoding artifacts and grammar errors.
Aswin 3	"little swept and smell ... played in the sleep ... became to healthy"	Words are locally plausible but the event sequence and argument structure fail.

The shared pattern is a mismatch between local character fluency and story-level planning. Lower-temperature or nucleus sampling, repetition blocking, cleaner Unicode normalization, longer context, and multiple windows per story are directly testable remedies.

1.6 Evidence map

Evidence	Repository path / identifier
Chaitanya metrics	task1_llm/member_chaitanya/metrics_report.csv
Chaitanya checkpoint	task1_llm/member_chaitanya/checkpoints/full_run01/final.pt; hash in reproducibility manifest
Chaitanya raw log	reproducibility/raw_logs/task1_llm/chaitanya/full_run01.log
Aswin metrics	task1_llm/member_aswin/metrics_report.csv (full_run01 row)
Aswin executed notebook	task1_llm/member_aswin/src/task1_colab_all_in_one.ipynb
Aswin samples	task1_llm/member_aswin/outputs/generated_text/full_run01_samples.jsonl

2. Task 2 - Yelp polarity sentiment classification

All six counted models use embeddings learned from scratch and are evaluated on the same stratified 5,000-review test set (2,500 positive, 2,500 negative). Chaitanya reproduced Aswin's row indices and verified texts and labels, enabling paired tests. Preprocessing still differs, so cross-member gaps combine preprocessing and architecture effects.

2.1 Model lineup

Model	Architecture	Parameters	Key training choices
C: fastText bigram	Mean unigram + hashed-bigram embeddings -> linear	19.29M	128-d; AdamW 3e-3; batch 64; early stop; dropout 0.3
C: HAN	Word BiLSTM + attention -> sentence BiLSTM + attention	2.75M	128-d; AdamW 1e-3; batch 64; dropout 0.3; clip 1.0
C: Transformer	2-layer, 4-head, width-128 hand-written encoder; masked mean pool	2.94M	AdamW 5e-4; 5% warm-up; dropout 0.1/0.2; clip 1.0
A: mean-pool FFN	Mean pool -> Dense 128 -> ReLU -> dropout -> logit	2.42M	128-d; AdamW 1e-3; wd 1e-4; batch 64; <=5 epochs
A: BiGRU	1-layer BiGRU 128/dir -> max+mean pool -> Dense 128	2.67M	AdamW 5e-4; dropout 0.4; batch 64; <=5 epochs
A: TextCNN	Conv1D kernels 3/4/5 x 100 -> max pool -> Dense 128	2.59M	AdamW 1e-3; dropout 0.4; batch 64; <=5 epochs

2.2 Shared test-set metrics

Model	Acc.	Macro P/R/F1	Micro F1	Weighted F1	TN/FP/FN/TP	ROC / PR	MCC
C: fastText bigram	0.9294	0.9294 / 0.9294 / 0.9294	0.9294	0.9294	2334/166/187/2313	0.9786 / 0.9792	0.8588
C: HAN	0.9286	0.9291 / 0.9286 / 0.9286	0.9286	0.9286	2281/219/138/2362	0.9810 / 0.9819	0.8577
C: Transformer	0.9128	0.9130 / 0.9128 / 0.9128	0.9128	0.9128	2258/242/194/2306	0.9728 / 0.9750	0.8258
A: mean-pool FFN	0.9162	0.9162 / 0.9162 / 0.9162	0.9162	0.9162	2295/205/214/2286	0.9737 / 0.9741	0.8324
A: BiGRU	0.9194	0.9197 / 0.9194 / 0.9194	0.9194	0.9194	2334/166/237/2263	0.9756 / 0.9766	0.8391
A: TextCNN	0.9252	0.9253 / 0.9252 / 0.9252	0.9252	0.9252	2290/210/164/2336	0.9800 / 0.9804	0.8505

Model	Acc. 95% CI	Macro-F1 95% CI	MCC 95% CI	Brier	ECE	McNemar vs own baseline	Params	Time	Examples/s	Peak MB
C: fastText bigram	[0.9218, 0.9362]	[0.9218, 0.9362]	[0.8436, 0.8724]	0.0543	0.0137	reference	19.29M	6.1s	16,463	698
C: HAN	[0.9218, 0.9354]	[0.9217, 0.9354]	[0.8440, 0.8713]	0.0537	0.0131	stat 0.030, p 0.8629	2.75M	49.0s	2,040	387
C: Transformer	[0.9050, 0.9206]	[0.9049, 0.9206]	[0.8100, 0.8413]	0.0668	0.0337	stat 23.593, p 0.0000	2.94M	16.5s	7,264	1,535
A: mean-pool FFN	[0.9084, 0.9242]	[0.9084, 0.9242]	[0.8168, 0.8484]	0.0605	0.0112	reference	2.42M	32.9s	3,040	1,059
A: BiGRU	[0.9116, 0.9272]	[0.9116, 0.9272]	[0.8236, 0.8549]	0.0610	0.0190	stat 0.978, p 0.3226	2.67M	3207.0s	31	437
A: TextCNN	[0.9176, 0.9324]	[0.9176, 0.9324]	[0.8353, 0.8650]	0.0556	0.0177	stat 5.275, p 0.0216	2.59M	46.3s	2,158	1,015

Team selection: Aswin TextCNN

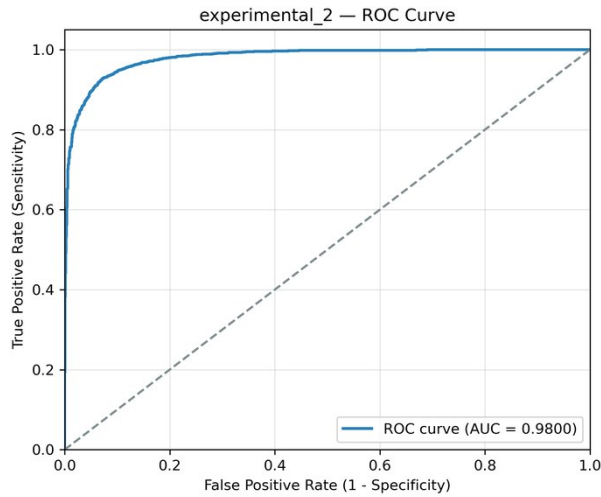
Aswin's TextCNN is the lead Task 2 model in this report: it is the strongest of Aswin's three runs (0.9252 accuracy, 0.9252 macro-F1, 0.9800 ROC-AUC) and uses only 2.59M parameters. Chaitanya's fastText-bigram baseline has a slightly higher point estimate (0.9294), but the paired difference is not significant after Holm correction (adjusted p = 0.9012). The TextCNN therefore gives the preferred accuracy/compactness trade-off while keeping Aswin's executed notebook and artifacts as the primary Task 2 evidence.

2.3 Slice robustness

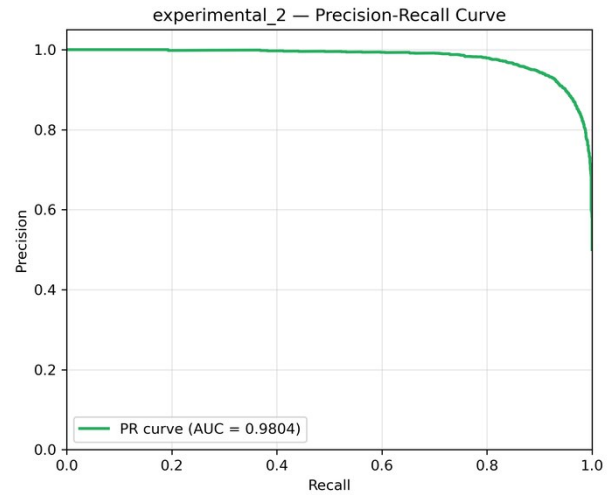
Model	Short F1/err	Medium F1/err	Long F1/err	Negation F1/err	Strong cues F1/err	Mixed F1/err	>5% UNK F1/err
C: fastText	0.922/0.075	0.933/0.067	0.925/0.072	0.928/0.069	0.958/0.041	0.923/0.077	0.901/0.094
C: HAN	0.926/0.072	0.934/0.066	0.919/0.080	0.926/0.073	0.956/0.043	0.921/0.079	0.904/0.090
C: Transformer	0.907/0.089	0.919/0.081	0.903/0.094	0.907/0.090	0.947/0.052	0.898/0.102	0.879/0.114
A: FFN	0.914/0.083	0.919/0.081	0.908/0.089	0.913/0.084	0.955/0.044	0.907/0.093	0.888/0.108
A: BiGRU	0.914/0.083	0.922/0.078	0.913/0.083	0.916/0.081	0.948/0.050	0.909/0.091	0.879/0.116
A: TextCNN	0.915/0.082	0.936/0.064	0.913/0.085	0.925/0.073	0.959/0.040	0.922/0.078	0.900/0.096

Cells show macro-F1 / error rate. Explicit sentiment cues are easiest. Mixed sentiment and high unknown-token rates expose the main weaknesses. TextCNN is the strongest Aswin model on medium, negation, strong-cue, and mixed slices; the local n-gram bias aligns with the bigram baseline's overall success.

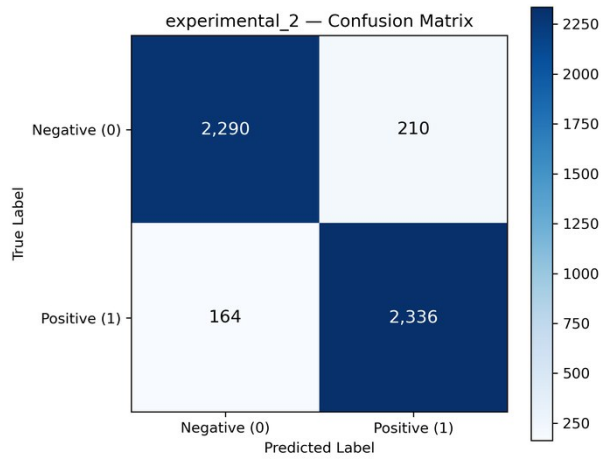
2.4 Aswin Task 2 evidence



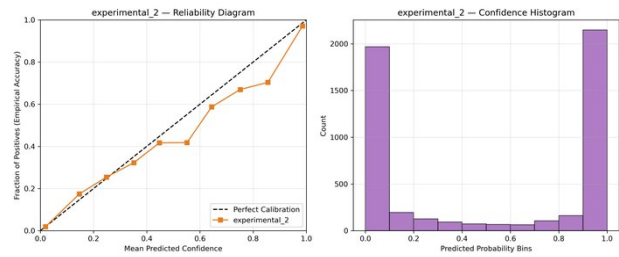
Aswin TextCNN ROC curve on the shared test split (ROC-AUC 0.9800).



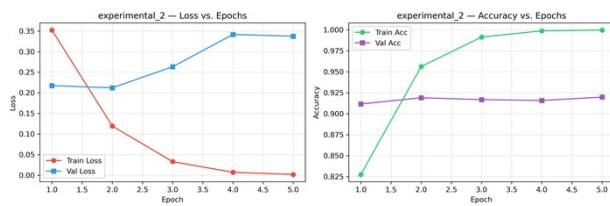
Aswin TextCNN precision-recall curve (PR-AUC 0.9804).



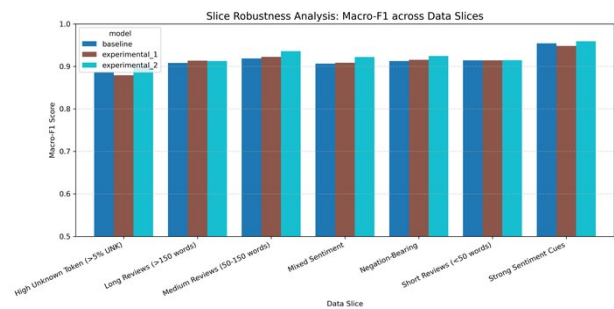
Aswin TextCNN confusion matrix: TN 2,290; FP 210; FN 164; TP 2,336.



Aswin TextCNN reliability diagram (Brier 0.0556; ECE 0.0177 in the member metrics file).



Aswin TextCNN training curves and validation behavior.



Aswin's seven-slice robustness comparison across the baseline, BiGRU, and TextCNN.

2.5 Joint error analysis

Member / category	Actual review excerpt	What failed / testable response
C: confident FP	"Average Japanese food at amazing Japanese food prices."	Sarcastic use of 'amazing'; add sentiment-incongruity or sarcasm features.
C: confident FN	"Sunday buffet for \$13 all you can eat and drink."	Positive label with no explicit sentiment cue; combine text with weak rating/context signals if allowed.
C: negation	"despite still not digging their ordering process, their food is just too good..."	Scope and double-negative structure; add last-sentence/verdict pooling and test on contrast slices.
C: mixed	"Very good staff ... The food is wretched ..."	Mean pooling sums conflicting cues; learn clause-level weighting.
A: confident FP	"Laser quest is a fun experience ... adults ... creep me out."	Mixed review dominated by positive vocabulary; add clause/position weighting.
A: confident FN	"Owner wrote me and asked me to actually give his business another chance..."	Subtle positive intent has no strong sentiment cue; the full audit proposes stronger discourse/context modeling.
A: near threshold	Long reviews with opposing clauses and late verdicts	Sequence truncation and weak conclusion weighting; hierarchical or last-sentence pooling is testable.
A: slice-specific	Reviews with negation or mixed sentiment	TextCNN local features help, but cannot always resolve which clause determines the label.

Aswin's 20-case audit is the primary Task 2 error analysis; Chaitanya's audit is retained for the required team comparison. Across both audits the dominant errors are mixed reviews, sarcasm, mild/factual sentiment, negation scope, and occasional label ambiguity. These are reasoning and discourse failures rather than ordinary vocabulary misses. Appendix A includes all 40 reviewed cases in compact form; the repository CSVs retain the complete text and proposed fixes.

2.6 Joint analysis and next steps

Aswin's TextCNN is the selected model because its convolutional filters preserve local 3-5 token expressions such as 'not good' while remaining compact. More generally, local phrase bias wins at this data scale: hashed bigrams and TextCNN filters preserve expressions that mean pooling loses, while the scratch Transformer lacks enough data to learn equivalent locality and is the least calibrated. The HAN's hierarchy does not improve mixed-review performance because many polarity reversals occur within a sentence. A fair next experiment would standardize preprocessing, train all models over three seeds, add conclusion-aware pooling, and measure paired McNemar changes on contrast, mixed, and unknown-token slices. The uncounted TF-IDF+logistic-regression reference reaches 0.935 accuracy, showing that strong sparse n-grams remain difficult to beat with only 20K training reviews.

2.7 Evidence map

Evidence	Repository path / identifier
Lead executed notebook	task2_sentiment/member_aswin/src/task2_sentiment_aswin.ipynb
Aswin metrics	task2_sentiment/member_aswin/metrics_report.csv
Aswin error review	task2_sentiment/member_aswin/outputs/error_analysis/error_analysis.csv
Aswin plots	task2_sentiment/member_aswin/outputs/plots/, outputs/confusion_matrices/, outputs/calibration/
Unified comparison	task2_sentiment/member_chaitanya/outputs/team_comparison.csv
Paired tests	task2_sentiment/member_chaitanya/outputs/team_mcnemar.csv
Slice table	task2_sentiment/member_chaitanya/outputs/slice_metrics.csv
Chaitanya metrics	task2_sentiment/member_chaitanya/metrics_report.csv
Chaitanya error review	task2_sentiment/member_chaitanya/outputs/error_review/baseline_candidates.csv

3. Task 3 - CycleGAN Monet/photo style transfer

A denotes Monet and B denotes photo. Every generated image used for evaluation is direct output from the members' own CycleGAN generators. The official comparison uses the instructor notebook on the first 300 sorted images in each real and generated folder; the score is $-(\text{FID} + \text{MiFID})/2$, so higher (less negative) is better.

3.1 Architecture and training comparison

Item	Chaitanya selected model	Aswin V2
Generators	ResNet-9, 64 filters, InstanceNorm, reflection padding, nearest-neighbor resize-conv	ResNet-9, 64 filters, InstanceNorm, reflection padding, nearest-neighbor resize-conv
Discriminators	A2B lineage single-scale; B2A specialization uses two-scale 70x70 PatchGAN	Single-scale 70x70 PatchGAN with InstanceNorm (no spectral normalization)
Objective	LSGAN; $\lambda_{\text{cycle}} 2$; $\lambda_{\text{identity}} 0$ in selected lineage	LSGAN; $\lambda_{\text{cycle}} 10$; $\lambda_{\text{identity}} 2.5$ decays to 0
Stability	50-image pool; DiffAugment; EMA 0.9999; weight averaging; per-direction selection	50-image pool; horizontal flip only (no DiffAugment); AMP; grad clip 10; EMA 0.999
Schedule	40-epoch base plus documented continuations to epoch 126; selected EMA 123	150 epochs; G lr $2e-4$, D lr $1e-4$; linear decay from 75; best epoch 130
Hardware	RTX 4090, fp32	RTX 4090, AMP

3.2 Official evaluation and Kaggle result

Member	A2B FID / MiFID	B2A FID / MiFID	Submission FID	Submission MiFID	Score	Leaderboard evidence
Chaitanya	97.806 / 0.4067	94.206 / 0.3928	96.0061	0.3998	-48.2029	Team submission; public -48.2029; team rank 15th
Aswin	109.561 / 0.4220	100.668 / 0.4054	105.1144	0.4137	-52.7641	Submitted under the team 2026-10-06; public -52.7640

Team selection: Chaitanya final model

Its official FID is 9.108 points lower and its MiFID is 0.0139 lower than Aswin V2, improving the competition score from -52.7641 to -48.2029. The documented progression from -53.83 to -48.20 came from DiffAugment, longer training, EMA, checkpoint averaging, weaker cycle/identity constraints, and direction-specific fine-tuning. The last 0.02-point gain is within snapshot noise and should not be overinterpreted.

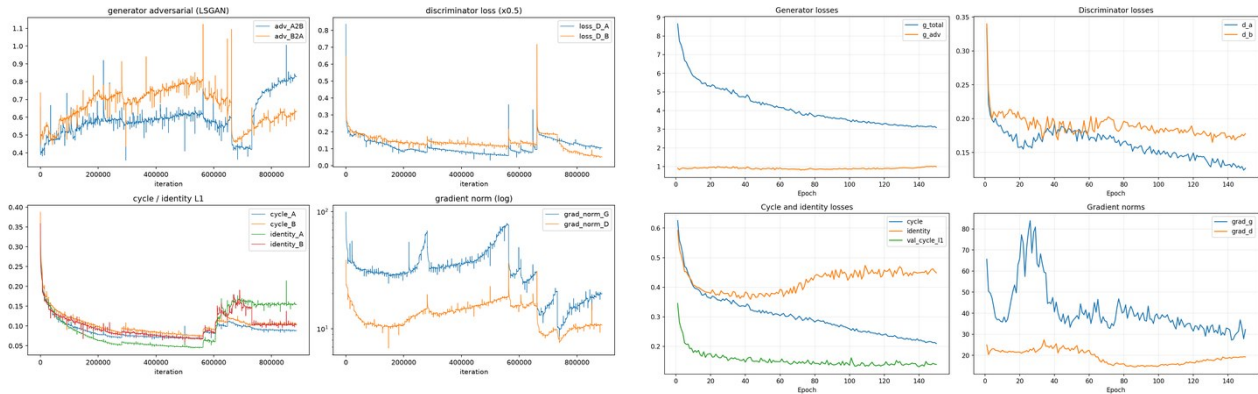
3.3 Full local metrics, both directions

Member / direction	FID	KID mean +/- std	Precision / recall	Density / coverage	Cycle L1	LPIPS	Content cosine
C A2B	81.8938	0.0211 +/- 0.0023	0.737 / 0.400	0.797 / 0.863	0.0445	0.4114	0.7390
C B2A	75.5026	0.0108 +/- 0.0023	0.597 / 0.690	0.611 / 0.840	0.0523	0.4412	0.7209
A A2B	92.5037	0.0238 +/- 0.0041	0.527 / 0.312	not computed	0.1101	0.4159	0.7716
A B2A	90.7133	0.0129 +/- 0.0034	0.425 / 0.583	not computed	0.1231	0.4630	0.7487

Local torchmetrics values and official notebook values differ because the feature implementations and sample protocols differ. Only the official notebook values are used for Kaggle selection; the local table is diagnostic. Chaitanya's final B2A model has higher recall and lower FID, while A2B has higher precision but lower recall. Aswin's B2A shows the same trade-off: recall 0.583 with precision 0.425.

3.4 Stability, losses, and efficiency

Item	Chaitanya final lineage	Aswin V2
Final G / D losses	G 1.849; D_A 0.052; D_B 0.105	Checkpoint G 3.214; D_A 0.129; D_B 0.179
Cycle / identity loss	A2B 0.088/0.103; B2A 0.106/0.154	Checkpoint cycle 0.229; identity 0.463
Gradient / non-finite	G=1824.527; D=381.466; NaNs 0	G mean 41.27; D mean 19.11; NaNs 0; skipped non-finite steps 70
Parameters	11,378,179 per generator	28,299,912 total; 11,383,427 per generator
Training time	31.8 h cumulative lineage	2.66 h
Throughput	A2B 204.2; B2A 303.0 images/s (generation)	16.92 train images/s
Peak GPU memory	18.3 GB	3.9 GB



Chaitanya run 18: generator, discriminator, cycle, and identity losses.

Aswin V2: generator, discriminator, and cycle losses over 150 epochs; no NaN losses, 70 skipped non-finite gradient steps.

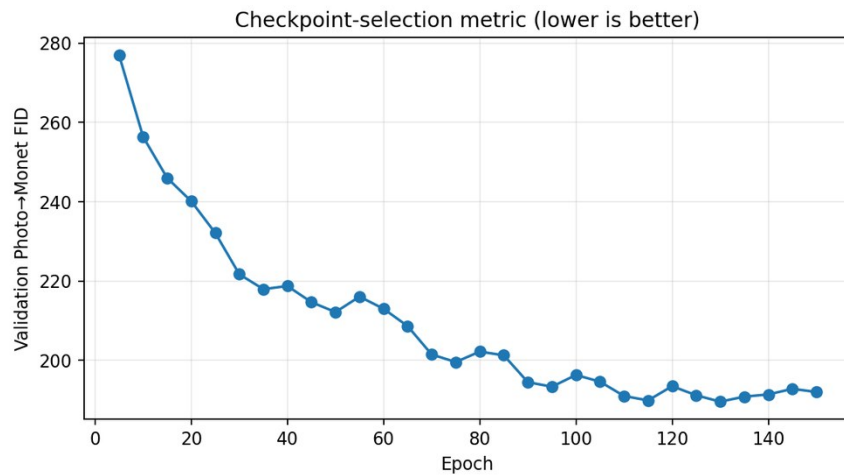


Figure. Aswin V2 validation B2A FID every 5 epochs; minimum 189.55 at epoch 130 selected `best_model.pt` (validation protocol, not the official one).
 task3_gan/member_aswin/outputs/plots/aswin_cyclegan_v2_validation_fid.png

3.5 Qualitative evidence and cycle consistency



Figure. Chaitanya selected checkpoint. Rows 1-3 show real Monet, Monet->photo, and cycle reconstruction; rows 4-6 show real photo, photo->Monet, and cycle reconstruction.

task3_gan/member_chaitanya/outputs/full_run18/grids/epoch_123.png



Figure. Aswin V2 fixed qualitative grid. The row order is printed in the image: real A, fake B, cycle A, real B, fake A, cycle B.

task3_gan/member_aswin/outputs/plots/aswin_cyclegan_v2_qualitative_grid.png

3.6 Failure analysis



Figure. Photo->Monet high cycle error: the faded pink input is recolored green/blue and reconstructs yellow-brown. Structure remains, but color information is discarded.

task3_gan/member_chaitanya/outputs/failure_cases/B2A_high_cycle_error/1_d0a1eed9dd.png



Figure. Photo->Monet lowest content similarity: an out-of-distribution night scene becomes daylight; point lights blob and skyline detail dissolves.

task3_gan/member_chaitanya/outputs/failure_cases/B2A_low_content/1_3a7a0992dd.png

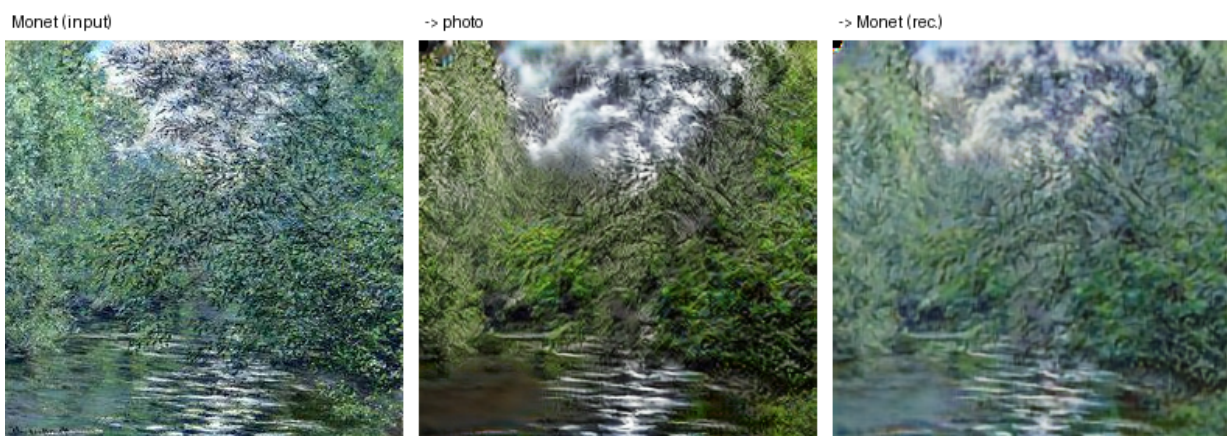


Figure. Monet->photo high cycle error: dense brushwork becomes streaky pseudo-photographic texture; signature and fine detail cannot be reconstructed.

task3_gan/member_chaitanya/outputs/failure_cases/A2B_high_cycle_error/1_4f7e01f097.png

Each strip is input | translation | reconstruction and was selected by a fixed metric-based ranking, not by visual cherry-picking. The cases show why low cycle error is necessary but not sufficient: the model can preserve reconstructable structure while hiding or discarding color and semantic details. Aswin's repository includes fixed audit samples but its failure_analysis.md has not yet been completed; the visible V2 grid suggests strong palette transfer with occasional saturation, texture smearing, and scene-detail loss, although raters scored V2 higher on artifacts than the submitted model (Section 3.7).

3.7 Blinded human audit (30 fixed inputs x 2 models, 2 raters)

30 inputs were fixed with seed 266 (20 photo-to-Monet, 10 Monet-to-photo; human_audit/inputs.txt). Each input was translated by both final models (Chaitanya run 18 EMA 123; Aswin V2 epoch 130), giving 60 items that were shuffled and shown under anonymous IDs with no model names. Both members rated every item independently on a 1-5 scale for style, content preservation, and artifacts (5 = no visible artifacts). The answer key (key.csv) was opened only after both raters had submitted.

Model / criterion	Mean	Rater 1 / Rater 2	B2A / A2B mean	Cohen's kappa (unweighted / quadratic)	Exact / within-1 agreement
Chaitanya: style	4.57	4.73 / 4.40	4.72 / 4.25	0.167 / 0.262	50.0% / 96.7%
Chaitanya: content	4.62	4.77 / 4.47	4.62 / 4.60	-0.182 / -0.198	40.0% / 96.7%
Chaitanya: artifacts	4.32	4.07 / 4.57	4.20 / 4.55	-0.184 / -0.279	23.3% / 73.3%
Aswin V2: style	4.70	4.67 / 4.73	4.72 / 4.65	0.088 / 0.250	63.3% / 96.7%
Aswin V2: content	4.65	4.80 / 4.50	4.62 / 4.70	0.267 / 0.267	63.3% / 100.0%
Aswin V2: artifacts	4.68	4.77 / 4.60	4.62 / 4.80	0.008 / -0.092	56.7% / 93.3%

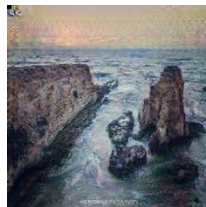
Model / criterion	Mean	Rater 1 / Rater 2	B2A / A2B mean	Cohen's kappa (unweighted / quadratic)	Exact / within-1 agreement
All 60 items: style	-	-	-	0.119 / 0.245	56.7% / 96.7%
All 60 items: content	-	-	-	0.040 / 0.017	51.7% / 98.3%
All 60 items: artifacts	-	-	-	-0.096 / -0.204	40.0% / 83.3%

Paired by input	Mean diff (Chaitanya - Aswin)	Chaitanya higher / Aswin higher / tie	Wilcoxon p
Style	-0.13	9 / 12 / 9	0.259
Content	-0.03	9 / 11 / 10	0.735
Artifacts	-0.37	3 / 19 / 8	0.002

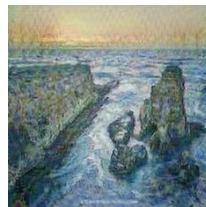
Both models score above 4.5 on style and content, and the paired differences there are not significant (p 0.26 and 0.74). The one significant difference is artifacts: Aswin V2 is rated cleaner on 19 of 30 inputs (4.68 vs 4.32, $p = 0.002$). This is the opposite of the FID ranking, which favours Chaitanya's model by 9 points: the looser cycle constraint that moves outputs closer to the Monet distribution also introduces visible artifacts. Inter-rater agreement is low (pooled quadratic kappa 0.25 style, 0.02 content, -0.20 artifacts) even though 83-98% of ratings fall within one point; with most scores at 4-5 there is little variance for kappa to credit. A shared rubric with anchor images is the clearest fix. A first round that rated only Chaitanya's model was superseded because the rater could identify their own model (archived in `human_audit/round1_own_model_only/`).



ca62c55ebe
(photo->Monet)
input



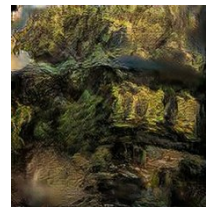
Chaitanya output



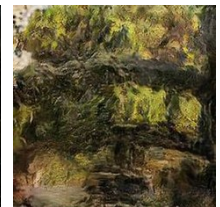
Aswin V2 output



d1d9748a64
(Monet->photo)
input



Chaitanya output



Aswin V2 output

`task3_gan/member_chaitanya/outputs/human_audit/images/` (items S01, S05, S08, S09; model mapping in `key.csv`; ratings in `rater_1.csv`, `rater_2.csv`)

3.8 Joint analysis and shortcomings

The best model improved the official distribution score by relaxing content constraints and specializing each direction. This is effective for FID but introduces a real trade-off: lower `lambda_cycle` and zero identity weight permit stronger style movement while weakening color and content preservation. EMA and weight averaging improve snapshot stability without changing inference architecture. The failed self-attention/spectral-normalization trials and non-transferable fine-tunes show that adding capacity or stabilization can unbalance the adversarial game. Aswin V2 is much cheaper and completes 150 epochs in 2.66 hours with about 4 GB peak memory, but its stronger cycle constraint (`lambda_cycle` 10) likely keeps outputs closer to inputs, and its discriminators train on un-augmented images: with only 300 paintings the Monet discriminator can memorise the training set, which DiffAugment counters (DiffAugment was Chaitanya's largest single gain, -53.87 to -51.47). Both factors plausibly explain V2's weaker official distribution match, but they were not ablated separately. Future work should pre-register a held-out model-selection subset, ablate one factor at a time, rate with a shared anchored rubric, and test a discriminator-scale ablation with identical schedules.

Open item: private leaderboard

Both submissions and the blinded two-model human audit are recorded. The private leaderboard scores become available only after the competition closes; add them here then.

3.9 Evidence map

Evidence	Repository path / identifier
Official team evaluation	task3_gan/member_chaitanya/outputs/official_eval.json and submission.csv
Chaitanya metrics	task3_gan/member_chaitanya/full_metrics_report.csv
Chaitanya design lineage	task3_gan/member_chaitanya/DESIGN_LOG.md
Chaitanya checkpoint	task3_gan/member_chaitanya/checkpoints/full_run18/epoch_123_ema/ (local, not in Git; source in outputs/pred_source.txt)
Human audit	task3_gan/member_chaitanya/outputs/human_audit/ (key.csv, rater_1.csv, rater_2.csv, audit_results.json); superseded round 1 in round1_own_model_only/
Aswin official evaluation	task3_gan/member_aswin/src/Part3_Evaluation_Script_evaluated.ipynb and submission.csv
Aswin metrics	task3_gan/member_aswin/full_metrics_report.csv
Aswin checkpoint	task3_gan/member_aswin/checkpoints/aswin_cyclegan_v2/best_model.pt (epoch 130)

4. Reproducibility and professional practice

Requirement	Evidence / status
Named member folders	Present for both members under all three task directories.
Executed notebooks	Preserved in each member's src folder; Task 3 official evaluation notebooks include outputs.
Metrics and analysis	CSV metrics, results.md, failure_analysis.md, plots, predictions, and checkpoint identifiers are linked above.
Raw logs and manifests	Committed under reproducibility/raw_logs and reproducibility/manifests where available; Aswin run logs are also retained in member output folders.
Smoke-test entry points	Top-level README documents smoke-test commands and artifact locations.
Personal paths / secrets	Final report uses repository-relative evidence paths; no credential is embedded.
Known incomplete evidence	Private leaderboard results (after competition close).

4.1 Threats to validity

- Task 1 split, cleaning, exposure, vocabulary, and distinct-n definitions differ; model-size conclusions are confounded.
- Task 2 preprocessing and seed counts differ across members; architecture is not the only changed variable.
- Task 3 repeatedly selects checkpoints on the same fixed official images; reported gains can be optimistically biased and small differences may be noise.
- Hardware and precision differ, so wall-clock comparisons measure full systems rather than architecture alone.

4.2 Final synthesis

Across all tasks, the strongest results come from matching inductive bias and evaluation to the data. More context and more training exposure improve the character GPT; local n-gram structure dominates sentiment at 20K examples; and EMA plus direction-specific specialization improves CycleGAN distribution matching. The compact alternatives remain important: Aswin's GPT and CycleGAN deliver large efficiency gains, and Aswin's selected TextCNN is statistically indistinguishable from the highest-scoring sentiment model under the paired team comparison. The final team recommendation is therefore score-led but evidence-aware: report the selected models, retain both members' independent artifacts, and present efficiency, confounds, and failure behavior alongside headline metrics.

Appendix A - Complete Task 2 error-review index

These compact tables include all 20 reviewed errors per member. Excerpts are shortened only for layout; complete unedited review text, model-visible tokens, explanations, and proposed fixes remain in the cited CSV files.

A.1 Chaitanya fastText-bigram baseline

#	Category	True/pred	P(pos)	Slices	Actual review excerpt
1	Confident FP	0/1	0.9996	short (<=60 words)	Average Japanese food at amazing Japanese food prices.
2	Confident FP	0/1	0.9972	short (<=60 words); no strong polarity words	Second trip to the bar v ased on a promo of free play. Gambling was good, great Pandora music, and plenty of tv. Bar keeps are to much chatty kathys among themselves.
3	Confident FP	0/1	0.9948	short (<=60 words); explicit rating mention	How do they get 5 stars? My fianclu00e9 was disappointed as well. The funkiest looking pad Thai ever.
4	Confident FP	0/1	0.9937	short (<=60 words)	I would recommend you the potatoes soup or clam chowder they are awesome... services is ok, kind of slow during lunch time
5	Confident FP	0/1	0.9926	short (<=60 words); contrast; no strong polarity words	Looks good from far but far from good.
6	Confident FN	1/0	0.0000	short (<=60 words); no strong polarity words	Decent!
7	Confident FN	1/0	0.0002	short (<=60 words); no strong polarity words	Sunday buffet for \$13 all you can eat and drink.
8	Confident FN	1/0	0.0019	short (<=60 words); negation; no strong polarity words; explicit rating...	I've just been forced to concede that, despite still not digging their ordering process, their food is just too good to disrespect with a 2 star review.
9	Confident FN	1/0	0.0020	short (<=60 words); contrast; no strong polarity words	Its a nice buffet looking over the Aria pool, but variety was just ok.... Meh..
10	Confident FN	1/0	0.0069	short (<=60 words); negation; no strong polarity words	If you don't \"get it,\" I'm \"sorry.\" Peel Pub... I shall forever *heart* you. _C\$
11	Near-threshold	1/0	0.4974	long (>180 words); negation; contrast	Let's be honest. Last Chance is not a place to take your date, your parents, or anyone else you're trying to show a good time. Yes, there are shoes and shirts strewn about - there are hun...
12	Near-threshold	1/0	0.4967	short (<=60 words); contrast; no strong polarity words	A unique park-like setting, located at the back of a modest farm. Many lovely things to look at while you wait. The brunch is served very casually, on paper plates, which seems strange fo...
13	Near-threshold	1/0	0.4966	medium (61-180 words); negation	I was astounded by the service and the food at this joint. Honestly, for \$150 per person including wine/cocktails/private dining room/tour of the kitchen/all the food we could eat in Vega...
14	Near-threshold	1/0	0.4965	medium (61-180 words); negation; no strong polarity words	This is a refreshing change from your small quick dining options while in Vegas. You're usually either stuck with the run of the mill 24 hour hotel lobby diner, a subpar casual dining exp...
15	Near-threshold	0/1	0.5042	medium (61-180 words); negation; contrast; no strong polarity words	We go here late night around 2 am every night we get out of the club nearby. If it weren't for the fact it's 2 am and we're usually pretty hammered I would never stick around for that lon...
16	Slice-specific (no strong polarity words, macro-F1 0.893, n=1918)	0/1	0.9894	short (<=60 words); negation; no strong polarity words; explicit rating...	The bull ride is fun. Very good staff -- they had great energy and really got people involved. The food is wretched -- worthy of a bio hazard warning. If it were not for the great staff a...
17	Slice-specific (no strong polarity words, macro-F1 0.893, n=1918)	0/1	0.9883	short (<=60 words); no strong polarity words	good: it's cheap, loud, every now and then a good band will play. good drink selection. bad: the parking and the crowd. it's fun to watch the young kids marvel in this new bar they've jus...
18	Slice-specific (no strong polarity words, macro-F1 0.893, n=1918)	1/0	0.0153	short (<=60 words); no strong polarity words	Owner wrote me and asked me to actually give his business another chance, said he was a new owner and proprietor that had taken over.
19	Slice-specific (no strong polarity words, macro-F1 0.893, n=1918)	1/0	0.0177	short (<=60 words); contrast; no strong polarity words	This is a very well run airport, but the security line can get way too long. The food choices in the terminals is just ok.
20	Slice-specific (no strong polarity words, macro-F1 0.893, n=1918)	0/1	0.9774	short (<=60 words); no strong polarity words	Well it's over when they want it to be underwear night 11-to whenever they feel it's over

A.2 Aswin best-model audit

#	Category	True/pred	P(pos)	Error type	Actual review excerpt	Proposed fix
1	Confident FP	0/1	0.9989	Sarcasm / Ironic Sentiment	Laser quest is a fun experience for kids of all ages, but the adults that play at this establishment, honestly kind of creep me out. The best time to br...	Augment training with punctuation/sentiment incongruity features or train an auxiliary sarcasm detection head.
2	Confident FP	0/1	0.9967	Domain Lexical / Contextual Mismatch	Second trip to the bar v ased on a promo of free play. Gambling was good, great Pandora music, and plenty of tv. Bar keeps are to much chatty kathys amo...	Add subword/character n-gram embeddings or character-level convolutions to handle domain terms and typos.
3	Confident FP	0/1	0.9965	Domain Lexical / Contextual Mismatch	Average Japanese food at amazing Japanese food prices.	Add subword/character n-gram embeddings or character-level convolutions to handle domain terms and typos.
4	Confident FP	0/1	0.9959	Contrastive Clause / Negation Scope	My Husband and I went for lunch last Saturday in Mesa and I can say that I am not a huge fan. I walked in to no electricity in the dining area. Stood in...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
5	Confident FP	0/1	0.9949	Domain Lexical / Contextual Mismatch	I would recommend you the potatoes soup or clam chowder they are awesome... services is ok, kind of slow during lunch time	Add subword/character n-gram embeddings or character-level convolutions to handle domain terms and typos.
6	Confident FN	1/0	0.0048	Contrastive Clause / Negation Scope	Last Friday when we arrived in Vegas we were hungry and wanted to find a place to eat lunch. We were staying at the Vdara and decided to walk across the...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
7	Confident FN	1/0	0.0056	Mixed Polarity Trade-off	This is a very well run airport, but the security line can get way too long. The food choices in the terminals is just ok.	Utilize a self-attention layer to model inter-token interactions between sentiment adjectives and topic nouns.
8	Confident FN	1/0	0.0060	Mixed Polarity Trade-off	Its a nice buffet looking over the Aria pool, but variety was just ok.... Meh..	Utilize a self-attention layer to model inter-token interactions between sentiment adjectives and topic nouns.
9	Confident FN	1/0	0.0063	Contrastive Clause / Negation Scope	I had dinner here last weekend, and decided to order the gnocchi since I had it at Bouchon in Yountville. I thought it was either overcooked and or dry,...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
10	Confident FN	1/0	0.0201	Subtle / Understated Positive	Owner wrote me and asked me to actually give his business another chance, said he was a new owner and proprietor that had taken over.	Increase vocabulary size to 40,000 or reduce min_freq from 3 to 2 to capture domain-specific positive modifiers.
11	Near-threshold	1/0	0.4986	Mixed Polarity Trade-off	Good sushi, good deal on the all you can eat. The service was a little slow but I'd go back again.	Utilize a self-attention layer to model inter-token interactions between sentiment adjectives and topic nouns.
12	Near-threshold	0/1	0.5015	Sarcasm / Ironic Sentiment	So as my girls and I were waiting 4 our table @ burGR, we decided to walk around and we were hungry and looking for a snack. We came upon this place. Lo...	Augment training with punctuation/sentiment incongruity features or train an auxiliary sarcasm detection head.
13	Near-threshold	0/1	0.5026	Near-Boundary Neutral Ambiguity	Yep, it appears closed for good.	Apply temperature scaling post-processing calibration or label smoothing during training to improve boundary margin.
14	Near-threshold	1/0	0.4942	Contrastive Clause / Negation Scope	a little confusing at times to get around but overall not too bad! Make sure to walk past the main food court to find a little more options. That being...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
15	Near-threshold	0/1	0.5070	Mixed Polarity Trade-off	Slow, disorganized, and it felt like they were more interested in selling glasses, contacts and all the add-ons than the health of my eyes. I've worn gl...	Utilize a self-attention layer to model inter-token interactions between sentiment adjectives and topic nouns.
16	Slice-specific (Negation / Mixed Sentiment)	0/1	0.7860	Contrastive Clause / Negation Scope	Well, we don't all have the same interests when it comes to pools. If you want to stare at better-than-average-in-Vegas bodies, you might really love th...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
17	Slice-specific (Negation / Mixed Sentiment)	1/0	0.0437	Contrastive Clause / Negation Scope	Harlow's gives me diarrhea. Every time. Every single time. You know what, though? I don't care because Harlow's cuts my hangover directly in half and I...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
18	Slice-specific (Negation / Mixed Sentiment)	1/0	0.4150	Contrastive Clause / Negation Scope	Called on Tuesday for a special occasion on Friday. Briefly explained to reservation-taker that I was surprising my husband for lunch on his 50th birthd...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
19	Slice-specific (Negation / Mixed Sentiment)	0/1	0.8278	Contrastive Clause / Negation Scope	Olives at the Bellagio overlooking the fountains we sat inside 8 tonight and after our standard have a drink at the bar moment we proceeded to the table...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...
20	Slice-specific (Negation / Mixed Sentiment)	1/0	0.3145	Contrastive Clause / Negation Scope	Okay, I wanted to give Pinot Brasserie only 3 stars for the insulting escargot but because of how well they treated us I bumped them up to four. I ate h...	Incorporate a positional attention weighting mechanism or dependency-parse clause weighting to give higher importance...

References

- Vaswani, A. et al. (2017). Attention Is All You Need. NeurIPS 30. arXiv:1706.03762.
- Eldan, R., and Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? arXiv:2305.07759.
- Zhu, J.-Y., Park, T., Isola, P., and Efros, A. A. (2017). Unpaired Image-to-Image Translation Using Cycle-Consistent Adversarial Networks. ICCV.
- Yang, Z. et al. (2016). Hierarchical Attention Networks for Document Classification. NAACL-HLT.
- Joulin, A. et al. (2017). Bag of Tricks for Efficient Text Classification. EACL.
- Zhao, S. et al. (2020). Differentiable Augmentation for Data-Efficient GAN Training. NeurIPS.
- Zhang, X., Zhao, J., and LeCun, Y. (2015). Character-level Convolutional Networks for Text Classification. NeurIPS.
- Holtzman, A. et al. (2020). The Curious Case of Neural Text Degeneration. ICLR.

Repository

https://github.com/ChaitanyaUppalapati/Lab_Pair_18

All numerical claims in this report are backed by the repository paths shown in the evidence tables.